Two Dietary Flavonoids "Beat" Acarbose by Docking and Lose by ~1 kcal/mol of Noise: A Vina Benchmark of Human Pancreatic Alpha-Amylase Inhibitors on 1XCX

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Abstract

Diabetes affected an estimated 828 million adults in 2022, including 36 million in Pakistan, where the prevalence among women rose from 9.0% in 1990 to 30.9% in 2022, the largest increase of any country, and most post-meal glucose control still depends on slowing starch digestion. Acarbose, the standard alpha-amylase and alpha-glucosidase inhibitor, blunts postprandial glucose but causes flatulence in 74% of patients, diarrhoea in 31% and abdominal pain in 19%, which limits adherence; dietary flavonoids are proposed as gentler adjuncts, but their published enzyme inhibition has rarely been benchmarked against acarbose in the same protocol. We ran the LeadForge pipeline live on 7 November 2026 (UniProtKB → ChEMBL → PubChem → RDKit → PAINS/ADMET → Vina docking): target human pancreatic alpha-amylase (UniProt P04746, AMY2A; ChEMBL CHEMBL2045, 23 bioactivities), structure PDB 1XCX at 1.9 Å, grid centred on the co-crystallised acarbose-derived ligand ([5.01, 16.45, 41.56]), and six ligands docked with AutoDock Vina 1.2.7 under an identical protocol. Luteolin scored −8.42 kcal/mol and quercetin −8.31 kcal/mol against acarbose at −7.35 kcal/mol, margins of 1.07 and 0.96 kcal/mol, inside Vina's ±1–2 kcal/mol noise, while both flavonoids carry PAINS catechol alerts and quercetin's oral bioavailability is under 10%. A ChEMBL acarbose isomer with a measured Ki of 12 nM scored −8.14 kcal/mol, and a second analogue (Ki 75 nM) scored −7.47 kcal/mol, ranking in the same order as their measured potency. The falsifiable next step is a head-to-head in vitro alpha-amylase inhibition assay of luteolin against acarbose under identical conditions with redox controls, with a pre-registered rule that a docking margin under 2 kcal/mol predicts no measurable advantage; a measured IC50 for luteolin better than acarbose's under those controls would refute the noise interpretation.

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1. Introduction

Diabetes is one of the fastest-growing disease burdens in the world: the NCD Risk Factor Collaboration estimated 828 million adults living with diabetes in 2022, up by about 630 million since 1990, and 445 million adults aged 30 years or older (59%) received no treatment. Pakistan carries the fourth-largest national burden, about 36 million adults, and the rise in prevalence among Pakistani women — from 9.0% in 1990 to 30.9% in 2022 — was the largest increase recorded for any country. More than 95% of adult diabetes is type 2, where blunting the post-meal glucose rise is a central therapeutic goal.

Human pancreatic alpha-amylase (HPA, AMY2A, UniProt P04746) starts starch digestion in the small intestine, hydrolysing complex starches to oligosaccharides; the catalytic triad is Asp197 (the nucleophile), Glu233 and Asp300, with five binding subsites spanning the active-site cleft. Acarbose, a pseudo-tetrasaccharide from *Actinoplanes utahensis*, competitively and reversibly inhibits pancreatic alpha-amylase and the membrane-bound intestinal alpha-glucosidases, delaying

carbohydrate digestion and lowering postprandial hyperglycaemia. It is minimally absorbed and acts locally in the gut — which is also the source of its problem: in US placebo-controlled trials (1,255 patients, 50–300 mg three times daily), flatulence occurred in 74% of acarbose-treated patients, diarrhoea in 31% and abdominal pain in 19%, against 29%, 12% and 9% on placebo, because undigested carbohydrate ferments in the colon. The side effects are the mechanism, so they cannot be dosed away.

Dietary polyphenols, especially flavonoids, are the obvious candidate adjuncts: they are present in everyday foods, they inhibit pancreatic alpha-amylase in vitro, and a 2019 structure–activity study found luteolin the best of five tested flavonoids — better than acarbose itself as the positive control — with quercetin and hesperetin also active by mixed-type inhibition, while catechin and rutin showed no inhibition. But that comparison was made in one assay system, and the flavonoid literature almost never benchmarks against acarbose in the same docking protocol. The falsifiable hypothesis of this paper is that, docked against human pancreatic alpha-amylase under one identical protocol, the dietary flavonoids luteolin and quercetin beat acarbose by less than Vina's ±1–2 kcal/mol noise margin — that is, the apparent win is a measurement of the method's limits, not a proven advantage — and that any genuine signal will instead appear in the ranking of measured ChEMBL potencies against their docking scores.

2. Methods

2.1 Literature basis

Sources were searched on 7 November 2026: the daily literature digest at `~/workspace/research-team/digests/` (2026-11-07, 595 papers screened, 8 selected — none on alpha-amylase, so this topic rests on targeted searches), PubMed and web search, ChEMBL (EBI), PubChem PUG REST and RCSB PDB. Key sources: the NCD Risk Factor Collaboration's 2024 *Lancet* analysis for the diabetes burden figures; the FDA acarbose label (DailyMed) for the mechanism and the 74%/31%/19% adverse-reaction incidences; the 1XCX and 1B2Y crystal structures and their primary citations for the active site; the 2019 flavonoid structure–activity study for the luteolin-beats-acarbose in vitro finding; and the 2022 *Trends in Food Science & Technology* review for quercetin's oral bioavailability below 10%.

2.2 Molecular docking

The LeadForge pipeline was run live on 7 November 2026 (UniProtKB → ChEMBL → PubChem → RDKit → PAINS/ADMET → Vina). Target: human pancreatic alpha-amylase, UniProt P04746, gene AMY2A, *Homo sapiens*, 511 residues; ChEMBL target CHEMBL2045, 23 bioactivities retrieved. Twenty-four compounds passed the Morgan-fingerprint similarity screen against acarbose and were ranked by composite score (similarity × 40 + Lipinski × 20 + Veber × 10 + PAINS × 10 + tox × 20). All 24 are acarbose analogues — large oligosaccharides, not drug-like molecules: the top five span molecular weights 483–2,366 Da with 6–39 rotatable bonds.

Docking was performed with AutoDock Vina 1.2.7, using the LeadForge protocol: the same structure, grid centre, box (22.5 Å cube), exhaustiveness (8) and pose count (3) for every ligand; best of three poses reported. The structure was human pancreatic alpha-amylase, PDB 1XCX (1.9 Å; acarbose rearrangement products bound), with the grid centred on the centroid of the full co-crystallised ligand — chain B pseudo-pentasaccharide, residues AC1 + BGC + GLC, 44 atoms: [5.01, 16.45, 41.56]. The centre was verified against the PDB file: the acarviosine-core (AC1) centroid [9.96, 17.05, 43.65] sits 4.01 Å from the catalytic-triad centroid (Asp197/Glu233/Asp300), the combined centre 7.23 Å, and the 22.5 Å box covers the full −3 to +2 subsite cleft. As a cross-check, the acarbose complex 1B2Y (3.2 Å) gives the same geometry: its AC1 centroid is 3.88 Å from its catalytic triad. Receptor preparation used RDKit (Gasteiger charges, simplified AutoDock atom types); ligand preparation used meeko. Vina's scoring noise is approximately ±1–2 kcal/mol; differences smaller than this

cannot be treated as significant on scores alone.

Six ligands were docked: acarbose itself (QUERY, the reference drug), the two dockable pipeline leads (CHEMBL1230193, rank 2; CHEMBL3618489, rank 4), an acarbose isomer (CHEMBL1233515, rank 6, same molecular formula C25H43NO18 as acarbose, measured Ki 12.0 nM — the most potent compound in the ChEMBL set), and two literature flavonoids added for the dietary-adjunct angle, luteolin (PubChem CID 5280445, C15H10O6) and quercetin (PubChem CID 5280343, C15H10O7), with SMILES verified live against PubChem PUG REST on 7 November 2026. Two pipeline leads were NOT docked and are reported as not docked, not as zero scores: CHEMBL3616594 (rank 3, MW 1,901 Da, 31 rotatable bonds, Ki 14.3 nM) and CHEMBL3616595 (rank 5, MW 2,366 Da, 39 rotatable bonds, Ki 41.6 nM) are far beyond Vina's reliable sampling range at exhaustiveness 8, and 39 torsions exceeds Vina's 32-torsion handling.

One pipeline fault was caught before reporting, and is documented here because it is part of the result. The pipeline auto-picked PDB 5U3A, an apo structure of human pancreatic alpha-amylase whose only non-water HETATM groups are the N-terminal pyroglutamate PCA (residue 1) and a glycan NAG 501. Both are in the pipeline's `_CENTRE_EXCLUDE` list, but the `_ligand_center` fallback ("first group with ≥ 6 atoms") re-picked PCA, giving centre [5.86, 82.81, 131.04] — the N-terminus, 19.5 Å from the catalytic-triad centroid (9.09, 81.66, 150.49). The run was killed before Vina produced any output, so no wrong-centre scores exist; the pipeline results were regenerated without docking, and all docking reported here is the corrected-centre re-dock on 1XCX. The root cause — the fallback ignoring `_CENTRE_EXCLUDE` — is a pipeline fix still owed in `dock.py`; until it is fixed, every auto-picked centre on an apo structure must be audited by hand.

Acarbose itself failed on the first docking attempt: Vina hit its 900-second cap while four to five sibling Vina runs shared the machine. It was retried once on a quiet machine (zero other Vina processes) with the identical protocol and completed in about 8 minutes at −7.35 kcal/mol. The first attempt's timeout is reported, not hidden: even the reference drug is a hard docking target.

2.3 PAINS screen

Every test compound was screened with the RDKit PAINS filter (FilterCatalog, PAINS A/B/C). Acarbose and all four ChEMBL compounds are PAINS-clean. Both flavonoids carry the same alert — catechol_A(92), from the 3′,4′-dihydroxy catechol B-ring present in luteolin and quercetin. A catechol alert is reported, never hidden; it changes the validation design (redox controls), because catechols can produce apparent inhibition through redox cycling and aggregation rather than active-site binding.

2.4 ADMET and toxicity read-across

Lipinski/Veber descriptors come from RDKit. Acarbose violates Lipinski three times (MW 645.61, 14 hydrogen-bond donors, 19 acceptors, TPSA 329 Å², logP −8.72) and fails Veber; the ChEMBL analogues are the same (2–3 violations each). The flavonoids are drug-like by the same rules (luteolin MW 286.24, quercetin MW 302.24, one rotatable bond each, zero Lipinski violations) — which is exactly why they are interesting and exactly why their PAINS alert and bioavailability matter more than their docking scores. A ToxGuard Tox21 kNN read-across panel (12 assays) was run for the ChEMBL compounds: it returned no neighbours (max similarity 0.000) for every ChEMBL lead, so the panel is uninformative for them and no tox claim is made from it; only acarbose itself had neighbours (max similarity 0.681, highest flag NR-ER 0.385). No read-across panel was run for the two added flavonoids.

2.5 Honesty limits

What this paper cannot show: docking is not activity; Vina noise is ±1–2 kcal/mol; no experimental data are presented; no molecular-dynamics refinement, binding-free-energy calculation, or re-docking validation of the co-crystallised ligand was performed; a single crystal structure (1XCX, one HPA

conformation) was used; induced-fit effects are not modelled; and the two most potent ChEMBL compounds by measured Ki (14.3 nM and 41.6 nM) could not be docked at all, so the docking set is missing its strongest measured binders. The flavonoid comparison rests on two molecules chosen from the literature, not on a screen.

3. Results

3.1 Docking: the comparison table

Compound	Known activity (ChEMBL CHEMBL2045)	Best affinity (kcal/mol)	Poses (kcal/mol)	MW	Rot. bonds	Lipinski viol.	PAINS
Luteolin	— (dietary flavonoid; PubChem CID 5280445)	−**8.42**	−8.42, −8.03, −7.89	286.24	1	0	catechol_A
Quercetin	— (PubChem CID 5280343)	−**8.31**	−8.31, −7.81, −7.62	302.24	1	0	catechol_A
CHEMBL1233515	Ki 12.0 nM	−8.14	−8.14, −7.81, −7.61	645.61	9	3	clean
CHEMBL1230193	Ki 75.0 nM	−7.47	−7.47, −7.33, −6.83	483.47	6	2	clean
Acarbose (reference)	— (no activity recorded on this target in this ChEMBL pull)	−7.35	−7.35, −7.14, −7.06	645.61	13	3	clean
CHEMBL3618489	Ki 14.7 nM	FAILED (conformer_failed)	—	1,435.34	23	3	clean

All docking on PDB 1XCX (human pancreatic alpha-amylase, 1.9 Å), grid centre [5.01, 16.45, 41.56] (co-crystallised ligand centroid), box 22.5 Å, exhaustiveness 8, Vina 1.2.7, 7 November 2026. Not docked: CHEMBL3616594 (Ki 14.3 nM, MW 1,901, 31 rotatable bonds) and CHEMBL3616595 (Ki 41.6 nM, MW 2,366, 39 rotatable bonds).

The headline, stated honestly: both dietary flavonoids "beat" acarbose on docking — luteolin by 1.07 kcal/mol and quercetin by 0.96 kcal/mol — but both margins sit inside Vina's ±1–2 kcal/mol noise, so neither is a proven advantage, and both compounds carry the PAINS catechol alert that the reference drug does not. The acarbose isomer CHEMBL1233515, the most potent compound in the set by measured Ki (12.0 nM), also scores better than acarbose by 0.79 kcal/mol — again inside noise. The one clean concordance in the table is between the two docked ChEMBL analogues: the 12 nM compound scores −8.14 kcal/mol and the 75 nM compound −7.47 kcal/mol, ranking in the same order as their measured potency, a 0.67 kcal/mol spread for a 6.25-fold Ki ratio (which corresponds to about 1.08 kcal/mol of binding free energy at 298 K — the same direction, smaller than the thermodynamic expectation, and inside noise). Pose consistency does not rescue the flavonoid "win": luteolin's best pose (−8.42) is 0.39 kcal/mol clear of its second pose, quercetin's by 0.50, and acarbose's own three poses span only 0.29 kcal/mol — all three molecules are stable in their poses, and the ranking between them is still a noise-level difference.

Two failures are part of the result. Acarbose's first docking attempt timed out at Vina's 900-second cap under CPU contention and only completed on a quiet-machine retry — a 645 Da oligosaccharide with 13 rotatable bonds is a slow, hard docking target even for the reference drug. CHEMBL3618489

(MW 1,435 Da, Ki 14.7 nM) failed earlier, at ligand preparation: no conformer could be generated (`conformer_failed`), so it has no score at all. The pipeline's lead list is dominated by molecules Vina cannot handle — three of the top five by rank (MW 1,435–2,366 Da) produced no docking score by one failure mode or another — which is itself a finding about this target class: similarity-to-acarbose screening returns compounds too large to dock, and the dockable subset is a biased sample of the small end.

3.2 Comparison with published docking

A 2024 fluorescence/docking study of acarbose and quercetin reports acarbose binding energies of −7.6 to −7.7 kcal/mol across multiple studies, with quercetin among the stronger reported alpha-amylase inhibitors. Our in-protocol acarbose value of −7.35 kcal/mol is 0.25–0.35 kcal/mol less favourable than that published range — inside Vina noise, and consistent in direction with the flavonoids scoring better. No published Vina score for luteolin on 1XCX was found to benchmark against, which is a gap this paper fills: Vina 1.2.7 reference values for acarbose, two flavonoids and two ChEMBL analogues on 1XCX are now on record for others to reproduce, together with the two documented failures. The published in vitro finding stands alongside the docking rather than behind it: Martinez-Gonzalez et al. (2019) measured luteolin as the best flavonoid inhibitor of pancreatic alpha-amylase, better than acarbose, by mixed-type inhibition — a real experimental result that our docking neither confirms nor refutes, because a 1.07 kcal/mol docking margin cannot confirm a measured effect.

3.3 The standardisation / quality problem

Two things that could silently corrupt this result did not, and two things that no structure check can catch did. The grid centre was verified against the PDB file to 0.01 Å and sits on the co-crystallised ligand, 4.01 Å (AC1 core) from the catalytic triad — the cofactor/glycan-centring fault is absent here, because the faulty 5U3A auto-pick was caught and the run killed before any score was produced. The duplicate-entry fault seen in Papers #1 and #3 did not occur: no lead is acarbose under another ID (CHEMBL1233515 shares acarbose's formula but has a different connectivity and a measured Ki of 12.0 nM). What did occur: the two most potent measured binders in the set could not be docked at all, and the reference drug itself needed a quiet-machine retry — so the docking table systematically excludes the strongest measured compounds. The deeper problem is the flavonoid one: both "winning" compounds carry the catechol PAINS alert, and quercetin's oral bioavailability is under 10% with extensive first-pass metabolism to glucuronide and sulfate conjugates and free-aglycone plasma levels in the low nanomolar range. A docking win for a catechol flavonoid is the starting point of an artefact investigation, not a discovery.

4. Discussion

4.1 The reframe

The result reframes the paper from a discovery claim into a measurement of Vina's limits on this target — with one honest exception. There is no proven flavonoid win: luteolin and quercetin beat acarbose by about 1 kcal/mol, inside noise, carrying a PAINS alert each. The durable findings are the benchmark values (acarbose −7.35 kcal/mol on 1XCX, with full pose lists), the ChEMBL pair ranking in the same order as measured Ki, the two documented docking failures, and the grid-centre fault caught before it produced scores. The flavonoid angle survives as a question, not as an answer: the published in vitro result (luteolin better than acarbose) is real experimental data, and our docking is consistent with it in direction but too noisy to confirm it.

4.2 Why the "win" must not be over-read

Four reasons. First, the margins (1.07 and 0.96 kcal/mol) are inside Vina's ±1–2 kcal/mol noise. Second, both winners carry the catechol_A PAINS alert — the classic flavonoid artefact signature —

while the reference drug and the ChEMBL analogues are PAINS-clean. Third, the single-structure limit: one HPA conformation, no induced fit, no re-docking validation of the co-crystallised ligand. Fourth, the set is biased: the three largest, most potent measured compounds in the ChEMBL pull could not be docked at all, so the table compares acarbose against the small end of its own analogue family plus two hand-picked flavonoids.

4.3 The bioavailability limit — and why it is weaker than it looks for this target

Quercetin's oral bioavailability under 10% is usually quoted as the reason flavonoid enzyme inhibition never translates — and for a systemic target it would be decisive. For pancreatic alpha-amylase it is a weaker objection than it looks, because the enzyme acts in the lumen of the small intestine: a dietary flavonoid meets its target in the gut before absorption, first-pass metabolism and plasma exposure ever enter the picture. The honest statement is narrower: poor systemic bioavailability does not kill a luminal-target adjunct, but it does not save it either. What matters is the flavonoid concentration actually present in the intestinal lumen at the time starch arrives, whether it survives gastric transit intact, and whether the measured inhibition survives redox controls — none of which docking addresses. The bioavailability number therefore constrains any systemic claim (and any claim about the absorbed metabolites) while leaving the luminal dietary-adjunct hypothesis testable in vitro.

4.4 The PAINS problem is the experiment's design constraint, not a reason to stop

Both flavonoids carry catechol_A(92). Catechols oxidise to quinones, redox-cycle, and aggregate — all mechanisms that produce apparent enzyme inhibition without active-site binding. The validation assay must therefore be designed to expose artefacts: run the inhibition assay with and without a reducing agent (e.g. DTT) and catalase, add a detergent counter-screen (e.g. Triton X-100) to test for aggregation-based inhibition, and test the oxidised quinone form separately. If luteolin's inhibition survives all three controls at the same IC50, the catechol alert is a false alarm for this compound; if the IC50 shifts or vanishes, the docking "win" was an artefact. The ChEMBL analogues, being PAINS-clean, need no such controls — which is a genuine advantage of the oligosaccharide series over the flavonoids, and the reason CHEMBL1233515 (Ki 12 nM, –8.14 kcal/mol) deserves the same assay slot as luteolin despite scoring 0.28 kcal/mol worse.

4.5 Testable hypotheses

• Luteolin inhibits recombinant human pancreatic alpha-amylase with an IC50 within 2-fold of acarbose's under identical assay conditions, with the inhibition surviving DTT/catalase and detergent controls — a measured IC50 for luteolin better than acarbose's under those controls would refute the noise interpretation; loss of inhibition under redox controls would confirm the PAINS artefact.

• CHEMBL1233515 inhibits recombinant HPA with an IC50 within 2-fold of its ChEMBL Ki (12 nM) under identical conditions, and more potently than CHEMBL1230193 (Ki 75 nM) — the docking predicts the same potency order; a reversed order would refute the docking ranking.

• A Vina re-dock of the co-crystallised ligand from 1XCX reproduces its crystallographic pose within 2 Å RMSD (protocol self-validation; not yet run).

• Fixing the `_ligand_center` fallback in `dock.py` to skip `_CENTRE_EXCLUDE` groups makes the pipeline auto-pick a ligand-bound HPA structure (e.g. 1XCX) instead of the apo 5U3A — a pipeline fix to be verified, not assumed.

4.6 Proposed validation (university-lab feasible, 8–12 weeks)

Stage 1 (weeks 1–2): in vitro HPA inhibition assay (p-nitrophenyl-α-D-maltopentoside or starch–DNS method) for luteolin, quercetin, CHEMBL1233515 and acarbose, same plate, n = 3, with a pre-registered 2 kcal/mol decision rule and the rule that a docking margin under 2 kcal/mol predicts no measurable advantage. Stage 2 (weeks 3–6): artefact controls for the flavonoids — DTT/catalase redox controls, Triton X-100 detergent counter-screen, and a test of the oxidised quinone form; any

compound whose inhibition vanishes under controls is dropped. Stage 3 (weeks 7–12): simulated gastrointestinal stability of luteolin (gastric pH, pancreatic conditions) to test whether the luminal-exposure argument survives contact with digestive conditions, and, if Stage 1 shows any genuine advantage, a starch-tolerance test in a rodent model with acarbose as control. Total compound cost is the binding constraint; luteolin and quercetin are cheap dietary flavonoids, the ChEMBL analogues are research chemicals, and none may be presented as a treatment.

5. Limitations

• Docking only; no experimental data. 2. Both "winning" margins (1.07 and 0.96 kcal/mol) are inside Vina's ±1–2 kcal/mol noise and cannot be interpreted as real advantages. 3. Both flavonoids carry the PAINS catechol_A alert; redox/aggregation artefacts are not excluded. 4. No molecular-dynamics refinement, binding-free-energy calculation, or re-docking validation of the co-crystallised ligand was performed. 5. A single crystal structure (1XCX, one HPA conformation) was used; induced-fit effects are not modelled. 6. The two most potent measured binders in the ChEMBL set (Ki 14.3 nM and 41.6 nM) could not be docked, and a third (Ki 14.7 nM) failed at conformer generation — the docking set is missing its strongest measured compounds. 7. Acarbose's first docking attempt timed out at Vina's 900-second cap and only completed on a quiet-machine retry; the reference score itself carries that sampling caveat. 8. The pipeline's `_ligand_center` fallback bug is documented but not yet fixed in code; the corrected centre here was applied by hand. 9. Quercetin's bioavailability figure (<10%) is a literature value, not measured here, and no luminal concentration was measured. 10. The flavonoids were hand-picked from the literature, not screened — selection bias in the comparison set.

6. Conclusion

No dietary flavonoid in this set beats acarbose: luteolin and quercetin score better on Vina docking by about 1 kcal/mol, inside the noise margin, carrying a PAINS catechol alert each, and quercetin's oral bioavailability is under 10%. The honest reading is that the published in vitro finding — luteolin better than acarbose — is consistent with our docking in direction but unconfirmed by it, and the next experiment is a head-to-head alpha-amylase IC50 assay with redox and detergent controls and a pre-registered 2 kcal/mol rule, not more Vina docking. The most useful numbers in this paper are the benchmark values themselves: acarbose −7.35 kcal/mol on 1XCX, with the failures that came with it.

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Data availability

All docking scores reported here are in `hidden_files/dock-HP-005c-2026-11-07.json` (corrected-centre re-dock) and the pipeline results in `hidden_files/dock-HP-005-2026-11-07.json`. ChEMBL IDs: acarbose (query), CHEMBL1230193, CHEMBL3618489, CHEMBL1233515, CHEMBL3616594, CHEMBL3616595. Flavonoid SMILES verified via PubChem: luteolin CID 5280445, quercetin CID 5280343. Target: UniProt P04746 (AMY2A), ChEMBL CHEMBL2045. PDB ID 1XCX. Re-dock script: `health-papers-agent/redock_hp005.py`; acarbose retry:

`health-papers-agent/retry_acarbose_hp005.py`; pipeline-only re-run: `health-papers-agent/run_pipeline_only.py`. No additional data were generated.